What's God to an Analytical Thinker:

The Cognitive Path to Atheism

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Atheism is a lack of belief in religious deities or spiritual forces, and it can be acquired in several ways. One commonly discussed route to atheism is the cognitive route, which posits that overriding the cognitive mechanisms and biases that contribute to religious belief can lead to atheism. In the following paper I will discuss the theoretical frameworks of, evidence for, and critiques against analytical atheism. The results suggest that there is a relationship between cognitive ability and atheism; however, the complex interaction of external factors affecting this relationship need to be clarified. I conclude with a brief discussion on the implications of these findings on our understanding of the psychological discourse surrounding religion.

Theoretical Frameworks

People have several cognitive mechanisms that facilitate, develop and maintain religious belief. Some of these mechanisms occur automatically and implicitly, like the tendency to be sensitive to environmental stimuli that resemble patterns (van Prooijen et al., 2018) or sentient agents (van Elk et al., 2016). Other cognitive mechanisms are developed over time and depend on external factors: theory of mind (ToM), which allows us to assign mental states to supernatural agents, is observed only after five years of age (Jenkins & Astington, 1996), and the tendency to anthropomorphise supernatural agents varies across religious doctrines (Muslims tend to anthropomorphize God less than Hindus; Shtulman et al., 2019).

When cognitive biases that facilitate religion are constrained, it promotes a type of atheism referred to as cognitive atheism. Overriding cognitive biases may be either a natural, non-effortful occurrence or a conscious, effortful process – the present paper will focus on this latter kind of atheism. Nevertheless, it is important to explore the distinctions between

differently acquired forms of cognitive atheism under various theoretical frameworks to understand what analytical atheism is and is *not*. I refer briefly to two examples to make this clarification: first, in people with autism, lower belief in God is mediated by their impaired ToM (Norenzayan et al., 2012), an example of "mind-blind atheism" posited by Norenzayan & Gervais (2013); second, people living in secure, non-life-threatening environments have higher rates of disbelief (Norris & Inglehart, 2011) theorized to occur due to less need for pattern and agency detection (Barrett, 2004), referred to as "apatheism" by Norenzayan & Gervais (2013) or "environmental atheism" by Kalkman (2014).

Though both these cases, disbelief is in part due to the weakening of cognitive mechanisms that promote religion, it is not a deliberate decision on part of the people, but merely a by-product of genetic or cultural factors. Kalkman (2014) makes a simpler distinction between a reflective route and unreflective routes in cognitive atheism, under which the cases I previously mentioned are classified as unreflective routes which involve natural, automatic and unconscious deficits in cognitive biases that dissuade religious belief in people. This paper focuses on the contrasting reflective route of cognitive atheism.

Reflective atheism is the conscious and effortful overturning of cognitive biases in cultivating religious disbelief (Kalkman, 2014). It is also referred to as "analytical atheism" (Norenzayan & Gervais, 2013). The core assumption is that people are able to consciously overcome intuitive cognitive biases that contribute to religious belief through analytical thinking, which might lead them to logically question religion and perhaps even reject it. This hypothesis is called the 'intuitive belief hypothesis' (Yilmaz, 2021) because it assumes that religious belief is cognitively intuitive, as opposed to atheism which requires analytical thinking.

Evidence for Analytical Atheism

Analytical atheism relies on an analytical cognitive style (ACS) based on facts and logical reasoning (Type 2 cognitive processing), as opposed to an intuitive style which is driven by automatic cognitive biases (Type 1; Evans & Stanovich, 2013). The degree to which a person thinks analytically versus intuitively is often assessed using the Cognitive Reflection Test (CRT), wherein higher scores imply higher analytical thinking and the categories are considered mutually exclusive (Frederick, 2005). Pennycook et. al (2016) found that higher scores were significantly negatively correlated with religious belief, with agnostics and atheists scoring higher than religiously affiliated people (see Figure 1). The inverse relationship, that intuitive thinking styles predict religious belief, has also been shown (Study 1 in Baimel et al., 2021; Finley et al., 2015; Study 1 in Shenhav et al., 2012)

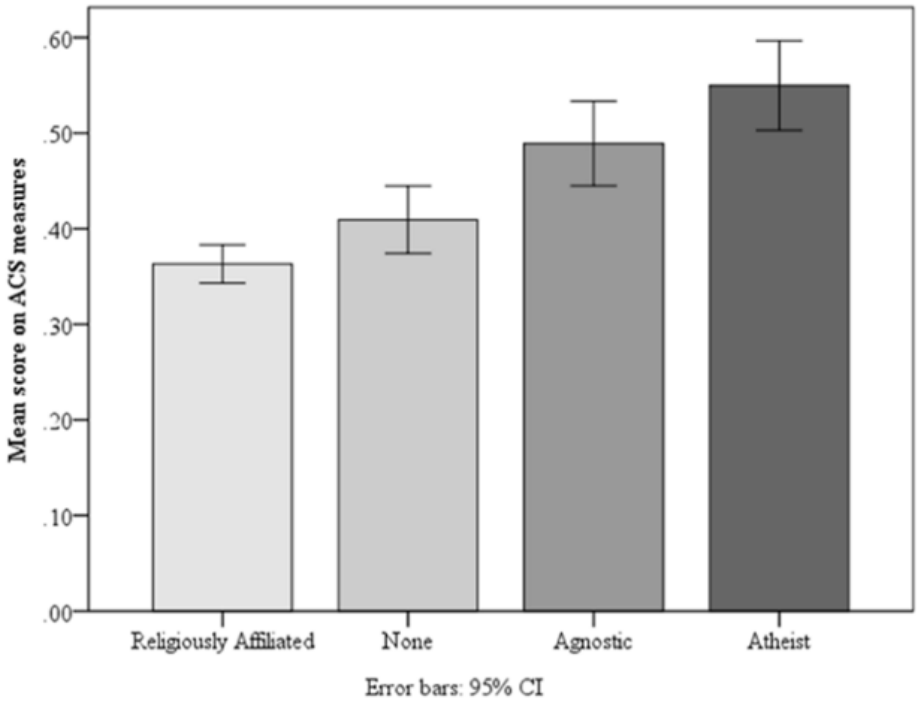


Figure 1. Accuracy on ACS measures as a function of religious affiliation as in Pennycook et al. (2016).

Da Silva and Carvalho (2015) found that atheists score higher than religious participants on the CRT; however, they also found that males score higher on the CRT than females,

suggesting that gender may play a role in this relationship. It also seems to be obvious that factors like intelligence and its predictors (level and type of education, age, parenting style, culture) would also systematically affect scores on the CRT. In this view, it may be argued that it is not analytical thinking itself that leads to atheism, but a person's gender, innate intelligence or amount of exposure to higher-ordering thinking. However, studies have shown that the relationship between analytical thinking scores on the CRT and atheism persists even after controlling for demographic factors like age, gender, political ideology and education (Pennycook et al., 2012). Recent studies show that the link between CRT and atheism exists across cultures and religions: participants from largely Judeo-Christian religions in the UK and majority-Hindu participants from India showed similarly negative correlations between religious belief and analytical thinking, controlling for political orientation and other demographic factors (Stagnaro et al., 2019); this link was also found to predict both intrinsic and extrinsic religiosity in a Turkish-Muslim population (Bahçekapili & Yilmaz, 2017).

It may be further argued that the *type* of education one has affects analytical thinking, specifically in religious thinking: psychologists or philosophers may be more prone to atheism by virtue of the arguments around religion in that field. Indeed, one of the earliest studies in the field showed that students of psychology and medicine held fewer paranormal beliefs than students of theology (Aarnio & Lindeman, 2005). To contend this hypothesis with relation to the discipline of philosophy, called the "reflection-philosophy confound", Byrd (2022) chose a sample of philosophers and non-philosophers alike and measured their CRT scores and religious attitudes. He found that CRT scores predict belief in supernatural agents like Santa Claus independent of philosophical training.

These findings do not sufficiently address the issue that higher analytical thinking may be

a consequence of general intelligence or other facets of cognitive processing. Researchers have questioned whether the CRT actually measures thinking style or only mathematical and logical capabilities (Campitelli & Gerrans, 2014). Further studies have shown that general cognitive ability and CRT scores differentially predict religiosity (Saribay & Yilmaz, 2017), and higher cognitive ability is also associated with religious disbelief (Razmyar & Reeve, 2013; Study 2 in Shenhav et al., 2012). Daws and Hampshire (2017) measured different facets of cognitive ability using a battery of tests in participants across 8 geographical regions – including the Middle East, South America, Asia, and Western countries. They found significant difference between the performance of atheists and agnostics as compared to religious participants, and that this relationship persists across factors including age, education and country of origin, providing cross-cultural evidence for atheism and cognitive ability (see Figure 2).

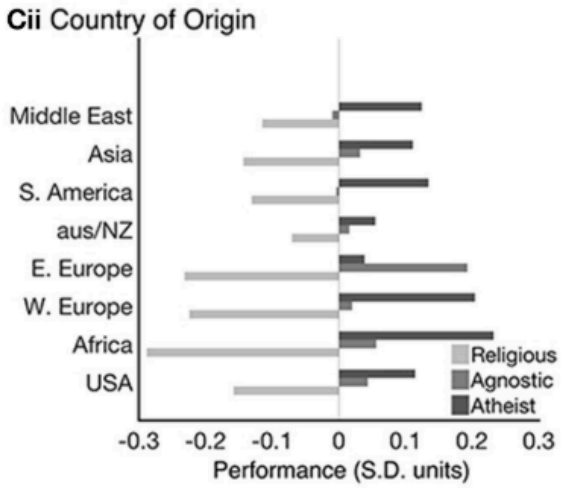


Figure 2. Overall performance on cognitive ability measures as a function of religious affiliation in various cultures as in Daws and Hampshire (2017).

The wealth of studies I have mentioned so far are only observational studies, and are thus limited in making a casual prediction about whether analytical thinking or cognitive ability can lead to disbelief. In other words, the studies only show a correlation between cognitive style and

atheism when controlled for other factors, but do not manipulate each construct systematically to observe the effect on the other. This problem, of course, exists in part due to the difficulty in manipulating religion or analytical thinking – one can hardly 'assign' a person to be religious or think analytically. However, certain experimental paradigms have attempted to manipulate thinking style by priming analytical or intuitive thinking, followed by a measure of religiosity in order to determine the casual relationship between the two, which I will discuss below.

Shariff et al. (2008) was one of the first studies to experimentally prime analytical thinking by using rational discourse on religion. The experimental condition participants were asked to read a passage by Dawkins that argued for secularism and respond with a short paragraph outlining their stance; the control condition involved writing about the neutral topic of one's favourite foods. Participants exposed to Dawkins' argument scored lower in both implicit and explicit measures of religiosity as compared to the control participants, revealing a tentative casual relationship between rational thinking and atheism. However, owing to the fact that Dawkins' passage directly addresses religion, there is a possibility that it is not the priming of analytical thinking but that of secularism that contributes to these results.

In Gervais and Norenzayan (2012) analytical thinking was primed implicitly and independent of religion. In Study 2, the experimental participants were exposed to pictures of sculptures that represented thinking (see Figure 3); in Study 3 and 4, the experimental group was given word sets that primed analytical thinking in a word-rearranging tasks; in Study 5, a hard-to-read font that was previously shown to trigger analytical thinking was used. All these studies included suitable, matched control conditions. The results showed that participants in whom analytical thinking was implicitly primed had increased disbelief in God and supernatural agents and lower implicit religiosity as compared to control participants. A replication of Study 3 in a

Muslim population showed similar results (Yilmaz et al., 2016).

Study 3 of Shehnav et al. (2012) manipulated which thinking style people favour by asking them to write a short paragraph talking about a situation in which their (1) intuition led to a good outcome (intuition-positive), (2) logic led to a bad outcome (reflection-negative), (3) intuition led to a bad outcome (intuition-negative) and (4) logic led to a good outcome (reflection-positive). The first two conditions primed intuitive thinking while the latter two primed analytical thinking. Results found that there was a significant difference in reported belief in God between participants who were primed to favour analytical thinking and those primed to favour intuitive thinking. However, it is important while interpreting these results to note that there was no control condition – such as asking participants to write a neutral paragraph – nor was there a pre-test to establish a baseline.

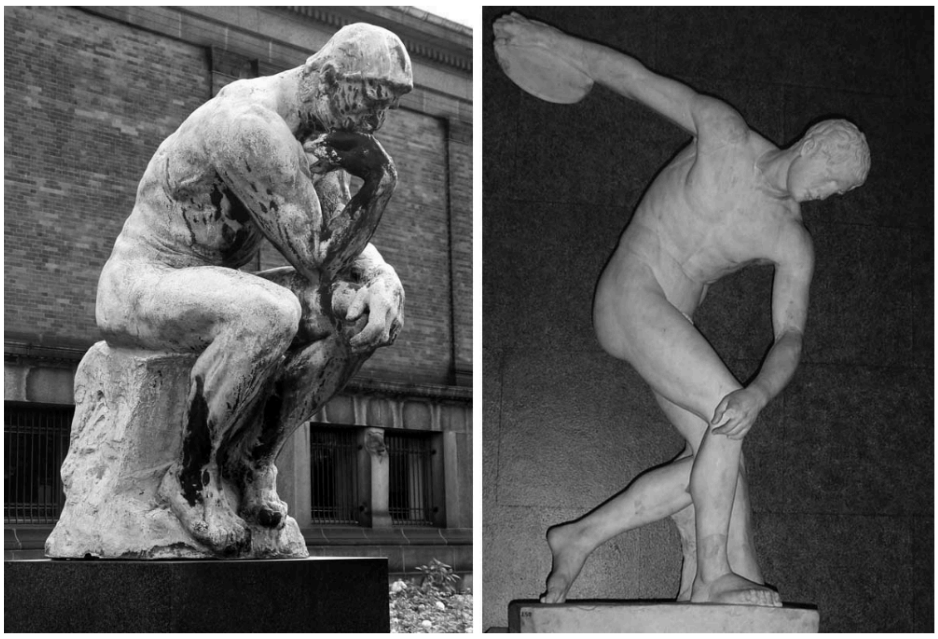


Figure 3. Priming stimuli in experimental condition (left) and control stimuli (right) in Study 2 of Gervais and Norenzayan (2012).

Analysing Analytical Atheism

Research in the field of analytical atheism suffers from issues of replication and generalizability – many of the findings previously mentioned were later found to have theoretical and empirical gaps. For example, the results of the experimental study conducted by Gervais and Norenzayan (2012) was not replicated in a subsequent study by Sanchez et al. (2017), and the authors of the original study themselves retracted confidence in the Study 2 findings due to the nature of the primes (see Figure 3 for recap; Gervais & Norenzayan, 2018). Gervais et al. (2018) were unable to find a clear link between CRT scores and atheism across a sample of 13 different countries. Shockingly, a moderate positive correlation between religious disbelief and analytical thinking was found in the UK, implying that analytical thinking actually *increased* belief in religion. However, it is important to note that the Stagnaro et al. (2019) paper mentioned in the previous section was conducted to directly address this study, and its results contradicted the findings of the Gervais et al. (2018) UK sample.

There are discrepancies within the literature of cognitive ability and thinking style in predicting belief: for example, in contrast with other literature, Razmyar and Reeve (2013) found that cognitive ability does predict religiosity but thinking style does not provide any unique, differential contribution to this relationship. A later commentary attributed this discrepancy to the lack of accuracy on the CRT within that particular participant pool, also acknowledging that studies involving the CRT are disadvantaged due the asymmetry between the three items on the CRT and longer and more comprehensive scales of cognitive ability (Pennycook, 2014). Further research by Farias et al. (2017) has found no significant effect of cognitive ability and thinking style on supernatural belief, when a pilgrimage site was used to prime religiosity (Study 1), working memory was loaded to deplete analytical thinking ability (Study 2) and brain stimulation was used to increase inhibition of intuitive thinking (Study 3). This points to a need

for more robust experimental paradigms to test whether thinking styles do, in fact, cause religious disbelief.

The lack of clarity in studying analytical atheism has often been attributed to the tendency to oversimply religious belief. Many critics have emphasized the importance of individual differences (Bahçekapili & Yilmaz, 2017), social norms (Caldwell-Harris et al., 2020), exposure to religious rituals or institutions (Gervais et al., 2021; Shults et al., 2018; Ståhl, 2021) and cultural environments (Daniel-Hughes, 2021; Weiss et al., 2021) as underlying factors in the relationship between cognitive performance and atheism. At the outset of this critical analysis, I would like to emphasize that the preceding paper has selectively reviewed research of what Kalkman (2014) termed the reflective route of cognitive atheism – however, a deeper understanding of atheism lies within the nuanced interaction of the unreflective pathways and external social, political, economic and cultural factors that a person.

Conclusion and Implications on Cognitive Science of Religion

To reiterate, the present paper relies on Kalkman's (2014) framework to focus on a reflective route of atheism that requires conscious effort in order to overturn natural cognitive mechanisms. This terminology is a consequence of discourse that surrounds atheism and religion, which has created binary labels of 'natural' and 'unnatural'. Many critics argue that atheism is unnatural because, as emphasized in the discipline of psychology of religion, humans have a natural tendency to develop religion in ordinary circumstances (Launonen, 2018). In a landmark paper, Geertz and Markússon (2010) posit that the claim that religion is natural is not deterministic but probabilistic. In other words, religion is natural only as a probability not a guarantee, hence atheism may also be natural. The naturality of religion does not exclude the naturality of atheism. However, critics assert that atheism flourishes only under specific cultural

circumstances which “upset” the natural development of theism (Barrett, 2010) and this external sociodemographic data is not a valid reflection of what may well be natural cognitive processes (Bering, 2010). There is a simple logic to these arguments – if atheism is natural, why does it selectively develop in Western, educated samples and not spontaneously across cultures?

The present paper shows mixed evidence in regards to the hypothesis that natural, intuitive thinking predicts religious belief. While some experimental studies have displayed that inducing analytical thinking does lower religious belief, broader questions of whether this type of thinking is effortful and persistent in everyday-life is yet unclear. Given that people may switch from one cognitive style to another in response to demands of a situation, it is obvious that environmental issues play a significant role in the development of atheism. However, there is a clear mediating effect of cognitive ability and thinking style in predicting atheism.

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